

Vaibhav Sharma

BACKEND & SYSTEMS ENGINEER // B.E. CSE · MITE · 2027

> sharma31stmay@gmail.com> +91 70107 63467> vaibhavsharmaa.me> github.com/sharmavaibhav31> linkedin.com/in/sharmavaibhav31
> Bangalore, India

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HERE

[IDENTITY]

CURRENT STATUS

Backend Engineering Intern · LazyStudents.in

DEGREE

B.E. Computer Science · CGPA 9.43 · Graduating 2027

CORE DISCIPLINE

Backend Systems · Event-Driven Architecture · Secure Engineering

OPEN SOURCE

Arachnode · 26 ★ · 47 forks · Project Admin (GSSoC)

LOOKING FOR

SDE-1 / Junior Backend Engineer · Product Companies · Startups · Full-time · Open to Relocation

[STORY]

I didn't start as a backend engineer. Like most first-year CSE students, I had no clear direction — just curiosity and a vague sense that I should be doing *something*. My first pull was toward cybersecurity: I got hands-on with Kali, explored ethical hacking, started a few courses. It was interesting, but I was consuming more than I was building.

Then I got pulled by the crowd toward mobile development. Did an internship, shipped code to a real app. But I noticed something uncomfortable: I was only productive when someone told me exactly what to do. I couldn't design a system from scratch. I was a passenger, not a driver. That bothered me enough to change direction.

Around the start of my third year, I started asking a different question — not "what should I build?" but "how do these things actually work?" Backend engineering and systems design clicked. Not because it was fashionable, but because it required *real* thinking: tradeoffs, constraints, failure modes, state. I also had a practical read on where AI was going — the hard parts of backend weren't being automated. So I doubled down on the hard parts.

I built projects from genuine need or curiosity, not to pad a portfolio. One turned into an open-source project with real contributors. The security interest never went away — my C shell with Seccomp sandboxing was me answering a question I had about running AI locally without exposing my system. Someday, maybe an MS in security. But right now, I want to build backend systems that don't fall over.

[EXPERIENCE]

Backend Engineering Intern

Jun 2026 — Present

LazyStudents.in · Remote

- Primary backend engineer on a plagiarism detection and reduction service — designing the text-comparison pipeline, backend API, and suggestion engine from scratch.
- Own end-to-end quality of **40+ platform tools**: writing structured test plans, identifying integration failures, and filing reproducible bug reports consumed directly by the backend team.
- Leading the interns team, coordinating task allocation and review cycles in an Agile environment.

AI/ML Backend Intern

Sep 2025 — Nov 2025

Infosys Springboard · Virtual

- Orchestrated 3 heterogeneous ML inference models (MusicGen, fine-tuned GPT lyric model, YAMNet genre classifier) behind a Flask API layer — MoodHarmonics: an end-to-end AI music generation system.
- Implemented model preloading and graceful fallback routing, cutting average initialization latency from ~2.2s to ~1.2s (**45% reduction**), measured over 500+ sequential requests.

→ Designed error-handling middleware ensuring inference continuity under model failures; persisted semi-structured metadata and hashed user credentials in MongoDB.

Flutter Developer Intern

Jul 2025 — Sep 2025

ClubChat · Remote · IIIT Delhi Startup

- Shipped event-feed and real-time chat to a production app with **8,000+ users**.
- Resolved **18 production stability issues** across two releases — including 3 critical crash-on-launch regressions — through log analysis and device-level reproduction.

[SYSTEMS BUILT]

Arachnode — Automated Job Discovery & Outreach Platform2026

PythonFastAPIRedis StreamsPostgreSQLDockerScrapyPlaywrightpytesttestcontainersOpen Source

Built because I genuinely needed it — manually tracking job postings and cold-emailing recruiters was unscalable. This system handles the discovery-to-draft pipeline automatically.

- Architected across **7 independent microservices** communicating via Redis Streams consumer groups — chose Redis Streams over Kafka for at-least-once delivery with lower operational overhead for a self-hosted deployment.
- **4-tier test suite**: unit, integration (testcontainers with ephemeral Postgres + Redis), contract (schema consistency across services), and e2e (full dashboard flows).
- OSINT recruiter enrichment (GitHub API + SMTP verification) and Ollama-powered cold-email drafting via Jinja2; full pipeline automated via APScheduler.
- Serves as Project Admin: reviewing PRs, enforcing API contracts, managing CI/GitHub Actions, mentoring GSSoC / ELUSOC contributors.

Stars 26Forks 47Commits 143Issues 32PRs Reviewed 25+

HPMS — Hostel Permission Management System2024–2025

Node.jsExpressPostgreSQLPrismaReactJWTDockerNginxn timeronQR CodesInstitution Project

Hostel students had to hand-write permission letters on A4 sheets — wardens were often unavailable, the process was slow and had no audit trail. I designed a full digital replacement.

- 5-role permission workflow (Student / Warden / Chief Warden / Admin / Security) with JWT-signed QR certificates generated on approval — valid for exactly 2 scans (exit + entry), auto-invalidated after both.
- Modular monolith backend (Node.js + Express + Prisma) across 7 feature modules: auth, users, master, requests, certificates, gate, admin — served behind Nginx reverse proxy with SSL termination for HTTPS-required camera access at the gate.
- Gate scanner web app using html5-qrcode; node-cron job detecting overdue returns and triggering warden alerts; SMTP notifications to parents on approval, exit, and return.
- Attempted Tatvik TMF20 biometric integration: built a .NET 4.8 bridge exposing the SDK over HTTP and a .NET 8 FingerprintService calling it — hit a firmware wall (SDK couldn't talk to third parties); pivoted to the QR-based flow as the production solution.
- Full SRS with 40+ FRs, ERD, and 4 system process flow diagrams; proposed to college administration for deployment.

Appraisal Management System2026

Java 21Spring Boot 4Spring SecurityPostgreSQLFlywayJWTDockerOpenAPI

The college's faculty appraisal process was entirely paper-based — no audit trail, no consistency. I built the backend to replace it.

- State-machine workflow: **DRAFT → SUBMITTED → HOD_REVIEWED → FINALIZED → LOCKED**, with legal revert paths and immutability post-finalization.
- Full audit trail on every transition (userId, timestamp, old/new value); JWT auth with RBAC across Faculty / HOD / Principal / Admin roles.
- Configurable scoring weights validated to sum to 100%; 13 Flyway migrations; @Valid on all endpoints; env-injected credentials via Docker Compose.

Academic Timetable Scheduling Engine2026

Java 17Spring Boot 3PostgreSQLFlywayH2 TestingApache POIDocker

Departments spend days generating timetables manually each semester. The constraint-satisfaction problem hooked me.

Greedy constraint-propagation engine enforcing **8 hard constraints**: no double-booking across faculty/room/section, 55-min consecutive-class

→ Greedy constraint propagation engine enforcing 6 hard constraints: no double booking across faculty; room-section; 60 min consecutive class gap; designation load caps; room-type matching; break isolation.

→ Section-first generation strategy prevents slot starvation; fallback faculty via priority-ranked FacultyPreference entries before failing.

→ Generates conflict-free timetables in **under 2 minutes** vs. days manually; H2 integration tests, Flyway V1-V10, Excel/PDF export, two-stage Docker build.

Offline AI Assistant Shell2025

Cllama.cppTinyLlama-1.1BSeccompLinux NamespacesrlimitsCMake

I wanted AI assistance in my terminal, locally, with no internet. Making it safe on a live system turned it into a real security engineering problem.

→ Sandboxed C shell integrating TinyLlama-1.1B via llama.cpp — fully offline, air-gapped execution post-setup.

→ Four-layer security: Seccomp syscall whitelist, PID/mount namespace isolation, capability dropping, rlimit caps — prevents privilege escalation and system exposure.

→ First-query latency ~2-3s (model load); subsequent ~1-2s; 10-20 tok/s CPU at ~1-1.5GB memory. Includes functional test suite, MD5 integrity check, and strace syscall audit workflow.

MoodHarmonics — End-to-End AI Music Generation System2025

PythonFlaskMusicGenYAMNetGPT (fine-tuned)PyTorchTensorFlowMongoDBReact

The goal wasn't to build another AI music app — it was to answer: can multiple ML models with different runtimes and constraints be orchestrated into a reliable, end-to-end system?

→ Orchestrated 3 heterogeneous ML models: MusicGen (text-to-audio), a fine-tuned GPT lyric model (enforcing verse/chorus structure vs. base GPT descriptive output), and YAMNet + GTZAN classifier (genre + confidence score) — all behind a single Flask API layer.

→ Model preloading at server startup + graceful fallback paths (local inference preferred, Hugging Face fallback) to manage heavy initialization and failure modes under real performance constraints.

→ Stored semi-structured song metadata and hashed user credentials in MongoDB; served generated audio from local file storage; coordinated frontend-backend interactions under CORS and latency constraints.

Air Notepad — Gesture-Based Drawing System2024

PythonOpenCVMediaPipeNumPyJupyter

Built for fun — wanted my own air-drawing app. Ended up at 176,000 LinkedIn impressions and 84 comments. Not every project needs to be enterprise-grade to matter.

→ Tracked 21 hand landmarks per hand via MediaPipe Hand Landmarker, mapping raw landmark coordinates to drawing strokes via weighted moving-average smoothing (configurable factor) to reduce jitter.

→ Dual-hand control: left hand for tool/color selection via pinch gesture (Euclidean distance threshold); right hand for drawing via index-finger extension detection using relative Y-coordinates.

→ HD canvas (1280×720), 3 tools (Pen/Brush/Eraser), 8-color palette; confidence thresholds at 0.8 for both detection and tracking; frame-by-frame NumPy array operations for minimal latency.

[TECHNICAL SKILLS]

LANGUAGES

Java · Python · SQL · C · JavaScript (Node.js)

BACKEND

Spring Boot · Spring Security · Spring Data JPA · FastAPI · Flask · Express.js · REST API Design · JWT · RBAC

DATABASES

PostgreSQL · Redis Streams · MySQL · MongoDB · Firestore · Flyway · Prisma ORM

DEVOPS & TOOLING

Docker · Docker Compose · Nginx · GitHub Actions · Linux · Bash · Git · Maven

TESTING

pytest · testcontainers · H2 (integration) · Unit / Contract / e2e

ARCHITECTURE
Event-Driven Microservices · Consumer Groups · Modular Monolith · API Gateway · State Machines · Audit Trails · Constraint Scheduling
SYSTEMS
Linux Namespaces · Seccomp · rlimits · llama.cpp · Syscall Analysis · .NET (bridge layer)
ML / CV
PyTorch · TensorFlow · MusicGen · YAMNet · MediaPipe · OpenCV · Ollama · Model Serving · Inference Optimization
FRONTEND
React · Vite · Flutter · Dart (mobile)

[CERTIFICATIONS]

ORACLE OCI 2025 DevOps Professional Nov 2025 · Verified
ORACLE OCI 2025 Data Science Professional Oct 2025 · Verified
ORACLE OCI 2025 Generative AI Professional Oct 2025 · Verified
ORACLE OCI 2025 Generative AI Foundations Associate Sep 2025 · Verified
GOOGLE CLOUD · COURSERA Cloud Security Engineer Specialization Jul 2024 · Verified
GOOGLE CLOUD · COURSERA Security in Google Cloud Specialization Sep 2024 · Verified
GOOGLE · COURSERA Google Cybersecurity Professional Certificate Verified

[EDUCATION]

B.E. Computer Science and Engineering Mangalore Institute of Technology and Engineering · Karnataka, India → CGPA: 9.43 / 10 → President, Coders Club — organized Hack n' Seek (700+ online · 180 on-site participants) → Mentored 60+ students through 6 technical workshops (Git, DSA, backend architecture)	Sep 2023 — Jun 2027
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[LEADERSHIP & COMMUNITY]

 Project Admin — Arachnode GSSoC · ELUSOC · Aperture 3.0 · Feb 2026 — Present Reviewing and merging contributor PRs, enforcing API contract standards, maintaining CI/GitHub Actions, coordinating releases for a project with 26 stars and 47 forks. Mentoring first-time open-source contributors. Merged 2 PRs into GitCanvas (Aperture 3.0) after maintainer review.	
 President, Coders Club — MITE Oct 2025 — Present	

Organized Hack n' Seek with **700+ online and 180 on-site participants** across multiple colleges. Delivered 6 technical workshops on Git, Data Structures, and backend architecture to 60+ students.

[WHERE I'M GOING]

PERSONAL_DIRECTIVE.log

I'm not entirely sure what the destination looks like yet, and I think that's honest. What I do know is the kind of engineer I want to be: someone who understands systems deeply enough to make real tradeoffs, not just implement tickets. I want to work on backend infrastructure where reliability is the product — distributed systems, data pipelines, secure architectures.

In 3–4 years: either a Senior Solutions Architect or a Senior Backend Engineer at a high-growth product company — or the person who helped a good startup go from 10 to 10 million requests a day. The cybersecurity interest is still live. An MS in Security is on the table.

The one thing I'm certain about: I want to build things that require real thinking. Things that break in interesting ways. Things where the decision I make about a message queue or a transaction boundary actually matters.